SymPy Bug Card

Bug 2: solveset claims $\sqrt{x^2} = x$ for all complexes

Task. Solve the principal-square-root equation over the complex numbers:

$$\sqrt{x^2} = x.$$

SymPy's answer.

$$\text{solveset}(\sqrt{x^2} = x,\ x,\ \mathbb{C}) = \mathbb{C}.$$

Correct answer.

$$\{z : \mathrm{Re}(z) > 0\} \cup \{it : t \in \mathbb{R},\ t \geq 0\},$$

in particular not $\mathbb{C}$.

Explanation. The set contains $x = -1$, yet $\sqrt{(-1)^2} = 1 \neq -1$. Squaring treats $\sqrt{x^2} = x$ as branch-insensitive, dropping the principal-root branch.

Likely root cause. In `sympy/solvers/solveset.py`, `_solve_radical` accepts the non-finite result produced after `unrad` turns the equation into $0 = 0$, without validating it against the original equation. This is a new instance of a known failure mode.

Artifact (GitHub).
`bug-report/bug-002-solveset-claims-sqrt-x-2-x-for-all-complexes/`